
Context-Aware Shot Value Estimation in the NBA Using Machine Learning

Nate Seluga and Michael O'Brien

Department of Computer Science

Harvey Mudd College

Claremont, California, USA

nsseluga@g.hmc.edu, micobrien@g.hmc.edu

Abstract

Traditional basketball metrics fall short in optimizing shot selection in the NBA. We use logistic regression and tree-based models based on player agnostic contextual features to predict shot success probability. Calibration analysis shows that all models produce well-aligned probability estimates, indicating reliable aggregate predictions despite high individual shot variability. Several visualizations further compare the efficiency of specific shots. Overall, this work demonstrates that context-aware modeling provides a meaningful approach for evaluating and optimizing shot selection in basketball.

1 Introduction

As sports like basketball adapt to incorporate modern technology, analysts look for computational advantages to find underlying factors that drive success. Outcome-based metrics like field goal percentage are no longer sufficient to evaluate shot quality because they do not account for context and the process of getting a shot. Prior work highlights the importance of using modern technology in finding advanced insights. *Applying Deep Learning to Basketball Trajectories* uses recurrent neural networks to try to understand how a ball's movement throughout a possession impacts shot accuracy, reinforcing the idea that dynamic aspects of a shot heavily impact its efficiency. *CourtVision: New Visual and Spatial Analytics for the NBA* incorporates spatial data to create new metrics "spread" and "range," helping to quantify the success of shots by location.

Our project expands on these ideas by building a machine learning model which quantifies shot quality based on player independent factors. Specifically, we aim to predict expected value of shots given a vector of contextual features to allow us to quantify which shots are inherently "good". Analyzing these models also gives us deeper insight into which factors are most important when determining shot quality.

2 Methods

We began with an exploratory data analysis to understand the relationship between pre-shot variables and shot outcome. This analysis guided decisions around feature selection and transformation. The dataset was then cleaned and preprocessed by removing missing shot clock values, normalizing time-based features and correcting shot types mislabeled by raw distance.

Shot location, shot distance, defender distance, number of dribbles, touch time and shot clock were the primary raw features from the dataset. We organized shots into bins for analysis based on type (ex. `three_pointer`) and creation (ex. `catch_and_shoot`). We also created new features regarding timing (ex. `time_pressure_index`), defensive pressure (ex. `defender_distance_ratio`) and nonlinear terms (ex. `shot_dist_squared`) to capture other complex relationships.

We modeled shot success as a binary outcome using multiple models to capture relationships of varying complexity. We began with logistic regression as a baseline linear model to provide interpretable probability estimates. To more directly account for nonlinear relationships and feature interactions, we implemented three tree-based models. The decision tree provides a simple nonlinear model. The random forest improves predictive performance by averaging across many trees to reduce variance. XGBoost further enhances this approach by sequentially learning from residual errors. Hyperparameter tuning used cross-validation to optimize log loss for each model. This ensured that each model was evaluated under well-calibrated and comparable settings. All models were trained on a training set and evaluated on a test set (80/20 split) using metrics like log loss and ROC-AUC to evaluate probability calibration and performance. Each of the four models was used twice: once using purely the raw data and once including the engineered features.

3 Results

Model	Raw Features			Engineered Features		
	Acc	ROC AUC	Log Loss	Acc	ROC AUC	Log Loss
Logistic Regression	0.6090	0.6300	0.6570	0.6113	0.6302	0.6563
Decision Tree	0.6163	0.6293	0.6532	0.6153	0.6283	0.6539
Random Forest	0.6199	0.6397	0.6487	0.6184	0.6346	0.6508
XGBoost	0.6191	0.6411	0.6477	0.6199	0.6365	0.6494

Table 1: Comparison of model performance using raw features versus engineered features.

Table 1 highlights that performance remains relatively consistent across models and feature sets, suggesting a natural limit to predictability in shot outcomes. While small gains are observed, particularly for logistic regression and XGBoost with engineered features, overall differences are modest. This reflects the inherent uncertainty in whether an individual shot is made or missed, as data cannot fully account for randomness in player execution. As a result, model performance appears to plateau across increasingly complex approaches. This inherent randomness is visualized in the appendix.

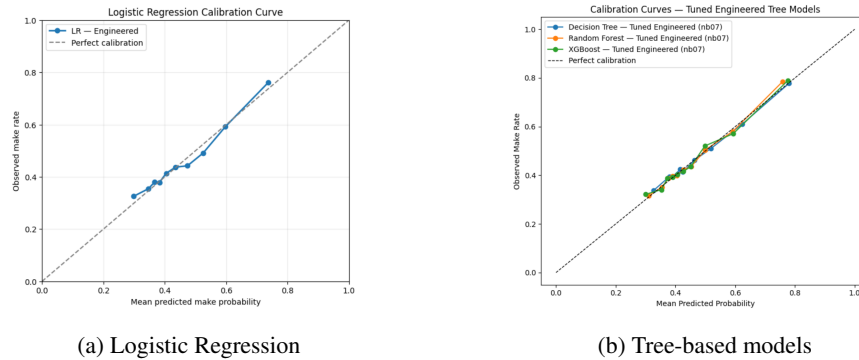


Figure 1: Calibration curves for all models using engineered features.

The calibration curves show that all models produce well-calibrated probabilities, with predictions closely aligning to observed make rates across most bins. This indicates that while individual shots remain inherently uncertain, the models reliably estimate average shot success when aggregated.

To better understand our model’s findings across different contexts we created three visualizations. Our primary visualization, pictured in Figure 2 below, is a heatmap displaying expected value for different combinations of defender pressure, dribbles and shot distance.

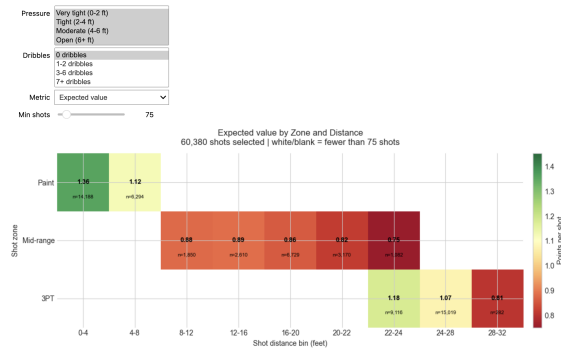


Figure 2: Expected value (points per shot) by shot zone and distance. Higher values indicate more efficient shots.

Several patterns emerge, namely the inefficiency of mid range shots compared to paint and three point shots, consistently having the lowest expected value. Corner threes, taken between 22-24 feet, are particularly efficient, consistently outperforming all other shots beyond 4 feet. The selection of shots shown in figure 2 focus on catch and shoot shots, where three pointers excel, however these trends were similar across all dribble counts.

Our additional visualizations, shown in the appendix, generally support these findings. Comparing shot value with defender distance across different shot types echoed the effectiveness of three point shots, indicating that their value may be less impacted by strong defense. Analyzing shot value compared to shot frequency similar denounced mid range shots, suggesting that they may be taken too often for their value. Overall, these results paint a clear picture of the shots that teams should try to create.

4 Conclusion

In this project, we developed a machine learning model which evaluates NBA shot quality based on contextual, player independent features. Our results demonstrate that these factors directly influence shot value. Further analysis of our model suggests that paint and three point shots hold more inherent value than mid range shots.

Our work is limited, as our model does not take into account valuable information like player skill as well as detailed spatial and movement data. Future work could expand on this idea of shot optimization by incorporating training based on team skill set, player movement and ball movement. Such additions could also help the model learn who should be taking which shots and how ball movement impacts a team’s ability to find efficient shots. Overall, our results suggest that a context aware analysis of shot value provides a strong baseline for understand and optimizing shot taking in modern basketball.

Contributions

Throughout the project, both collaborators were involved with every major research and modeling decision. We contributed collaboratively to each report, with each individual focusing on what they did each week and throughout the project.

Nate Seluga: I first headed the dataset management, finding a useful dataset for our project and cleaning mislabeled shot value tuples. I then created the baseline logistic regression model using the raw features provided by the dataset and determined their role in the model. I performed an evaluation to measure the overall effectiveness of our initial model and then engineered new relational features based on the evaluation for later models. I also added hyperparameter tuning to the final models to maximize accuracy. After our improved models were finalized, I helped evaluate their efficacy and possible improvement based on standard evaluation metrics and simple visualizations. From there, I created the analytical visualizations, helping us gain useful insights and demonstrate the model's findings in an easily understandable way.

Michael O'Brien: I first took the raw dataset and parsed through it to determine potentially useful features to be engineered from the provided dataset that could capture the raw context of a shot better and represent higher-level basketball concepts. I also conducted the exploratory data analysis to understand the distribution of key variables in the raw dataset and notice any potential patterns early on. I then created the initial tree-based models (decision tree, random forest, gradient boosted via XGBoost) and ran them all on the raw features provided by the dataset and determined the quality of each model as well as the highest impact variables. I then took the findings we gathered from these initial runs of the models plus the finalized engineered features list and reran the each model with the new features. I created the evaluation results from these findings and constructed a few simple visualizations to portray these findings. Finally, I made another visualization in which the inherent uncertainty of shot outcome was pictured compared to the overall trends captured.

References

Goldsberry, K. (2012). CourtVision: New Visual and Spatial Analytics for the NBA. MIT Sloan Sports Analytics Conference.

<https://www.sloansportsconference.com/research-papers/courtvision-new-visual-and-spatial-analytics-for-the-nba>

Shah, R., & Romijnders, R. (2016). Applying Deep Learning to Basketball Trajectories. arXiv preprint.

<https://arxiv.org/abs/1608.03793>

Appendix

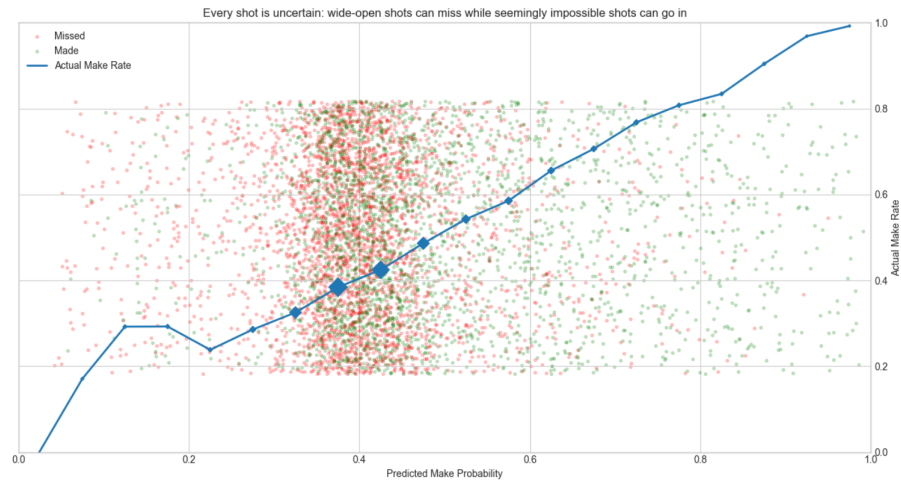


Figure 3: Predicted make probability vs. actual make rate, illustrating model calibration across all shots. Individual made (green) and missed (red) shots highlight the inherent uncertainty in shot outcomes.

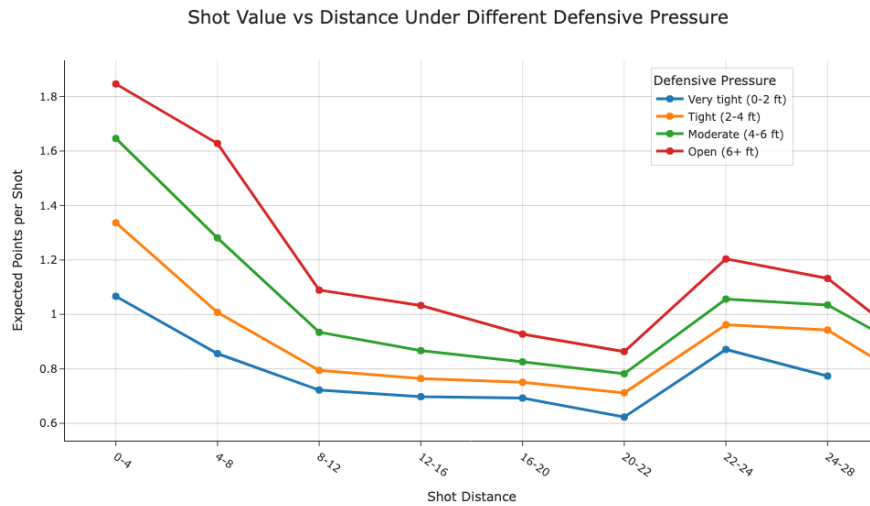


Figure 4: Expected value as a function of defender distance across different shot types.

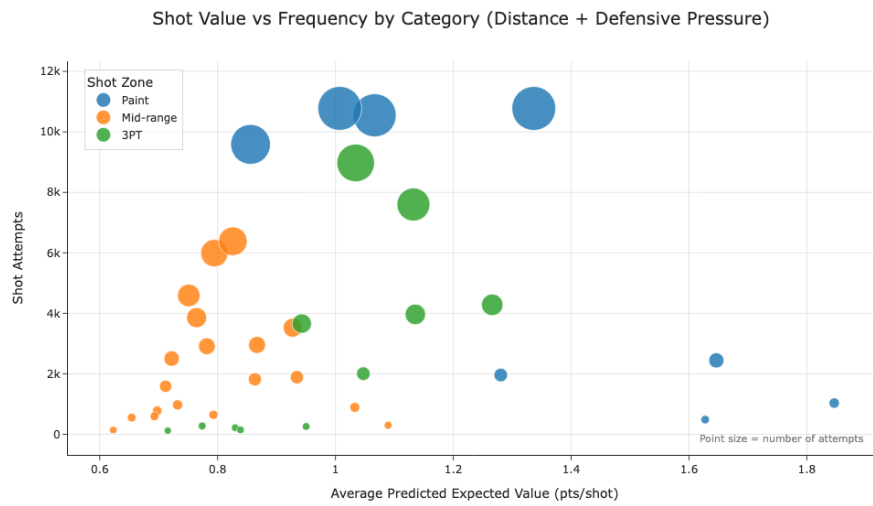


Figure 5: Shot value versus frequency by category (distance and defensive pressure).